1 Hybrid amortized configuration-conditional spectral calibration model

1.1 Campaign-level data, metadata, and notation

Let the training corpus be a collection of campaigns

$$\mathcal{D} = \{\mathcal{C}_m\}_{m=1}^{M_{\text{camp}}},$$

where campaign m is represented by

$$\mathcal{C}_m = (\mathcal{S}_m,\ K_m,\ f_m,\ c_m,\ Y_m).$$

Here:

- $\mathcal{S}_m \subseteq \{1, \dots, S_{\max}\}$ is the set of sensors participating in campaign m, with cardinality
$$M_m = |\mathcal{S}_m| \geq 2;$$
- $K_m \in \mathbb{N}$ is the number of aligned acquisitions in campaign m;
- $$f_m = (f_{m,1}, \dots, f_{m,J_m}), \qquad f_{m,1} < \cdots < f_{m,J_m},$$
is the common frequency grid retained after campaign preprocessing;
- $$c_m \in \mathcal{X}$$
is the acquisition configuration, assumed constant within campaign m:
$$c_m = (f_{\text{c},m},\ B_m,\ \text{RBW}_m,\ g_{\text{LNA},m},\ g_{\text{VGA},m});$$
- $$Y_m = \{Y_{m,s,k,j}\}$$
is the aligned PSD tensor in linear power units, indexed by
$$s \in \mathcal{S}_m, \qquad k \in \{1, \dots, K_m\}, \qquad j \in \{1, \dots, J_m\}.$$

Thus each campaign is internally fixed-config, but different campaigns may have different configurations c_m and different frequency grids f_m.

Ancillary metadata such as campaign label, date range, and acquisition cadence may be stored for bookkeeping, filtering, or optional side-information. However, the physically relevant observation operator is determined by c_m and the retained frequency grid f_m.

1.2 Forward observation model

For campaign m, sensor $s \in \mathcal{S}_m$, acquisition k, and frequency bin j, we posit

$$Y_{m,s,k,j} = G_{m,s,j}\, S_{m,k,j} + N_{m,s,j} + \varepsilon_{m,s,k,j}, \tag{1}$$

where

- $S_{m,k,j} \geq 0$ is the latent common spectrum for campaign m at acquisition k and bin j;
- $G_{m,s,j} > 0$ is the multiplicative calibration gain of sensor s in campaign m at frequency $f_{m,j}$;
- $N_{m,s,j} \geq 0$ is the additive floor of sensor s in campaign m at frequency $f_{m,j}$;
- $\varepsilon_{m,s,k,j}$ is the residual term.

Conditionally on (G, N, S), the working likelihood is Gaussian:

$$\varepsilon_{m,s,k,j} \mid G_{m,s,j}, N_{m,s,j}, S_{m,k,j} \sim \mathcal{N}\Big(0, \sigma^2_{m,s,j}\Big), \tag{2}$$

with campaign-, sensor-, and frequency-dependent residual variance $\sigma^2_{m,s,j} > 0$.

The corresponding calibrated sensor-domain estimate is

$$X_{m,s,k,j} = \frac{\big[Y_{m,s,k,j} - N_{m,s,j}\big]_+}{G_{m,s,j}}, \tag{3}$$

where $[x]_+ = \max\{x, 0\}$ or, in implementation, clipping to a small positive floor $\varepsilon_S > 0$ for numerical stability.

1.3 Hybrid amortized parameterization

The campaign-specific calibration parameters are not fitted independently from scratch. Instead they are generated by one global trainable model with parameters θ.

1.3.1 Sensor and campaign encoders

Each sensor identity $s \in \{1, \ldots, S_{\max}\}$ is associated with a trainable embedding

$$e_s \in \mathbb{R}^{d_s}.$$

The acquisition configuration is encoded by a trainable map

$$u_\theta : \mathcal{X} \to \mathbb{R}^{d_c}, \qquad u_m := u_\theta(c_m).$$

For each sensor $s \in \mathcal{S}_m$, define a sensor summary

$$r_{m,s} \in \mathbb{R}^{d_r}.$$

If sensor s is visible to the encoder, then

$$r_{m,s} = \phi_\theta(Y_{m,s,\cdot,\cdot}, f_m, c_m)\,, \tag{4}$$

where ϕ_θ is a sensor-wise encoder operating on the full time-frequency tensor of that sensor within campaign m. If sensor s is masked during training, then

$$r_{m,s} = r_\varnothing, \tag{5}$$

where $r_\varnothing \in \mathbb{R}^{d_r}$ is a learned mask token.

The campaign context is produced by a permutation-invariant aggregation of the visible sensor summaries:

$$h_m = \rho_\theta(\{(e_s, r_{m,s}) : s \in \mathcal{V}_m\}, u_m) \in \mathbb{R}^{d_h}, \tag{6}$$

where $\mathcal{V}_m \subseteq \mathcal{S}_m$ is the visible subset used by the encoder during training, and ρ_θ is permutation-invariant with respect to the sensor order.

1.3.2 Frequency-basis parameterization

To support campaigns with different frequency grids and to enforce smoothness with respect to physical frequency, the calibration curves are generated through basis expansions.

Let

$$\begin{aligned} b_\ell^\gamma &: \mathbb{R} \to \mathbb{R}, \quad \ell = 1, \dots, L_\gamma, \\ b_\ell^\eta &: \mathbb{R} \to \mathbb{R}, \quad \ell = 1, \dots, L_\eta, \\ b_\ell^\nu &: \mathbb{R} \to \mathbb{R}, \quad \ell = 1, \dots, L_\nu, \end{aligned}$$

be fixed frequency basis functions (for example, cubic B-splines defined over the relevant frequency range).

For each campaign m and sensor $s \in \mathcal{S}_m$, the model generates coefficient vectors

$$a_{m,s}^\gamma = A_\theta^\gamma(e_s, r_{m,s}, h_m, u_m) \in \mathbb{R}^{L_\gamma}, \tag{7}$$

$$a_{m,s}^\eta = A_\theta^\eta(e_s, r_{m,s}, h_m, u_m) \in \mathbb{R}^{L_\eta}, \tag{8}$$

$$a_{m,s}^\nu = A_\theta^\nu(e_s, r_{m,s}, h_m, u_m) \in \mathbb{R}^{L_\nu}, \tag{9}$$

where A_θ^γ, A_θ^η, and A_θ^ν are trainable hypernetworks.

These coefficient vectors induce unconstrained campaign-specific curves

$$\bar{\gamma}_{m,s,j} = \sum_{\ell=1}^{L_\gamma} a_{m,s,\ell}^{\gamma}\, b_\ell^{\gamma}(f_{m,j}), \tag{10}$$

$$\eta_{m,s,j} = \sum_{\ell=1}^{L_\eta} a_{m,s,\ell}^{\eta}\, b_\ell^{\eta}(f_{m,j}), \tag{11}$$

$$\nu_{m,s,j} = \sum_{\ell=1}^{L_\nu} a_{m,s,\ell}^{\nu}\, b_\ell^{\nu}(f_{m,j}). \tag{12}$$

1.3.3 Identifiability projection

The forward model (1) is invariant under the campaign-wise transformation

$$G_{m,s,j} \mapsto a_{m,j} G_{m,s,j}, \qquad S_{m,k,j} \mapsto a_{m,j}^{-1} S_{m,k,j}, \qquad a_{m,j} > 0.$$

Therefore an identifiability constraint is required.

Let $G_{m,s,j}^{(0)} > 0$ denote a nominal baseline gain. In the current project philosophy one may set

$$G_{m,s,j}^{(0)} \equiv 1.$$

We impose identifiability by projecting the unconstrained log-gain corrections to zero mean across the participating sensors:

$$\gamma_{m,s,j} = \bar{\gamma}_{m,s,j} - \frac{1}{M_m} \sum_{r \in \mathcal{S}_m} \bar{\gamma}_{m,r,j}, \qquad s \in \mathcal{S}_m,\ j = 1, \ldots, J_m. \tag{13}$$

Then the multiplicative gains are

$$G_{m,s,j} = G_{m,s,j}^{(0)} \exp(\gamma_{m,s,j}). \tag{14}$$

When $G_{m,s,j}^{(0)} \equiv 1$, (13) implies

$$\frac{1}{M_m} \sum_{s \in \mathcal{S}_m} \gamma_{m,s,j} = 0 \quad \Longleftrightarrow \quad \prod_{s \in \mathcal{S}_m} G_{m,s,j}^{1/M_m} = 1, \tag{15}$$

so the latent spectrum is defined on the campaign-specific network reference scale whose geometric mean gain equals one at each frequency bin.

1.3.4 Positive floor and positive variance

Define the softplus map

$$\mathrm{sp}(x) = \log(1 + e^x).$$

Then

$$N_{m,s,j} = \mathrm{sp}(\eta_{m,s,j}), \tag{16}$$

$$\sigma^2_{m,s,j} = \sigma^2_{\min} + \mathrm{sp}(\nu_{m,s,j}), \tag{17}$$

where $\sigma^2_{\min} > 0$ is a small fixed variance floor.

Thus the trainable model outputs campaign-specific calibration parameters

$$\{G_{m,s,j}, N_{m,s,j}, \sigma^2_{m,s,j}\}_{s\in\mathcal{S}_m,\ j=1,\dots,J_m}$$

through a single global parameter vector θ shared across all campaigns.

1.4 Explicit latent-spectrum update

The latent spectrum is not produced by a free black-box decoder. Instead, given the amortized campaign-specific parameters (G, N, σ^2), it is computed analytically by weighted least squares.

Define

$$\omega_{m,s,j} = \frac{1}{\sigma^2_{m,s,j}}. \tag{18}$$

For any subset $\mathcal{U}_m \subseteq \mathcal{S}_m$, define the subset-specific latent-spectrum estimator

$$\widehat{S}^{(\mathcal{U}_m)}_{m,k,j} = \left[\frac{\sum_{s\in\mathcal{U}_m} \omega_{m,s,j}\, G_{m,s,j}\, \left(Y_{m,s,k,j} - N_{m,s,j}\right)}{\sum_{s\in\mathcal{U}_m} \omega_{m,s,j}\, G^2_{m,s,j}}\right]_+. \tag{19}$$

At training time, to avoid trivial leakage into masked sensors, the latent spectrum is inferred from the visible subset:

$$\widehat{S}_{m,k,j} := \widehat{S}^{(\mathcal{V}_m)}_{m,k,j}. \tag{20}$$

At inference time on a new campaign, all available sensors are visible, so

$$\widehat{S}_{m,k,j} := \widehat{S}^{(\mathcal{S}_m)}_{m,k,j}. \tag{21}$$

The returned calibrated measurements are then

$$\widehat{X}_{m,s,k,j} = \frac{\left[Y_{m,s,k,j} - N_{m,s,j}\right]_+}{G_{m,s,j}}. \tag{22}$$

Thus the complete output of the model for campaign m is

$$\left(\widehat{X}_m,\ \widehat{S}_m,\ G_m,\ N_m,\ \sigma^2_m\right).$$

1.5 Self-supervised masked-sensor training objective

1.5.1 Visible and masked subsets

During training, each campaign sensor set is randomly partitioned into

$$\mathcal{S}_m = \mathcal{V}_m \,\dot{\cup}\, \mathcal{B}_m,$$

where

- $\mathcal{V}_m$ is the visible subset used by the encoder;
- $\mathcal{B}_m$ is the masked subset used for self-supervised prediction.

The encoder receives only the visible sensors, but the model must still predict calibration parameters for all sensors.

1.5.2 Masked negative log-likelihood

The primary self-supervised objective is the masked negative log-likelihood

$$\mathcal{L}_{\text{mask}}^{(m)} = \sum_{s\in\mathcal{B}_m} \sum_{k=1}^{K_m} \sum_{j=1}^{J_m} \left[\frac{\left(Y_{m,s,k,j} - G_{m,s,j}\widehat{S}_{m,k,j} - N_{m,s,j}\right)^2}{2\sigma_{m,s,j}^2} + \frac{1}{2}\log \sigma_{m,s,j}^2 \right]. \tag{23}$$

Because $\widehat{S}_{m,k,j}$ is computed from the visible subset only, the masked sensors cannot be reconstructed by direct copying. The model must infer transferable campaign structure from the visible sensors, sensor identities, and configuration metadata.

1.5.3 Optional visible reconstruction term

To stabilize optimization, one may include a visible reconstruction term

$$\mathcal{L}_{\text{vis}}^{(m)} = \sum_{s\in\mathcal{V}_m} \sum_{k=1}^{K_m} \sum_{j=1}^{J_m} \left[\frac{\left(Y_{m,s,k,j} - G_{m,s,j}\widehat{S}_{m,k,j} - N_{m,s,j}\right)^2}{2\sigma_{m,s,j}^2} + \frac{1}{2}\log \sigma_{m,s,j}^2 \right]. \tag{24}$$

The visible term is optional because it is easier and may dominate training if over-weighted.

1.6 Frequency smoothness regularization

Since $G_{m,s,j}$, $N_{m,s,j}$, and $\sigma^2_{m,s,j}$ are functions of physical frequency, smoothness should be imposed with respect to the actual frequency grid f_m, which may be nonuniform.

Let

$$\tilde{h}_m = \mathrm{median}(f_{m,j+1} - f_{m,j})_{j=1}^{J_m-1},$$

and define normalized coordinates

$$x_{m,1} = 0, \qquad x_{m,j} = \sum_{u=1}^{j-1} \frac{f_{m,u+1} - f_{m,u}}{\tilde{h}_m}, \qquad j = 2, \ldots, J_m.$$

For $r = 1, \ldots, J_m - 2$, set

$$h^-_{m,r} = x_{m,r+1} - x_{m,r}, \qquad h^+_{m,r} = x_{m,r+2} - x_{m,r+1}.$$

Define the nonuniform three-point second-difference operator $D_{m,2} \in \mathbb{R}^{(J_m-2)\times J_m}$ row-wise by

$$(D_{m,2}z)_r = \frac{2}{h^-_{m,r}(h^-_{m,r} + h^+_{m,r})} z_r - \frac{2}{h^-_{m,r}h^+_{m,r}} z_{r+1} + \frac{2}{h^+_{m,r}(h^-_{m,r} + h^+_{m,r})} z_{r+2}. \tag{25}$$

Then the smoothness penalty is

$$\mathcal{R}^{(m)}_{\mathrm{smooth}} = \sum_{s\in\mathcal{S}_m} \left[\lambda_\gamma \left\|D_{m,2}\gamma_{m,s,\cdot}\right\|_2^2 + \lambda_\eta \left\|D_{m,2}\eta_{m,s,\cdot}\right\|_2^2 + \lambda_\nu \left\|D_{m,2}\nu_{m,s,\cdot}\right\|_2^2\right], \tag{26}$$

where

$$\gamma_{m,s,\cdot} = (\gamma_{m,s,1}, \ldots, \gamma_{m,s,J_m})^\top, \quad \eta_{m,s,\cdot} = (\eta_{m,s,1}, \ldots, \eta_{m,s,J_m})^\top, \quad \nu_{m,s,\cdot} = (\nu_{m,s,1}, \ldots, \nu_{m,s,J_m})^\top.$$

The gain regularizer acts on log-gain corrections, not on G directly; likewise the additive floor regularizer acts on the softplus preimage η, not on N directly.

1.7 Optional reliable-sensor anchoring

If a campaign-level reliable sensor $s^\star_m \in \mathcal{S}_m$ is available from a separate ranking diagnostic, one may encourage that sensor to remain close to the nominal gain baseline through

$$\mathcal{R}^{(m)}_{\mathrm{anchor}} = \sum_{j=1}^{J_m} \gamma^2_{m,s^\star_m,j}. \tag{27}$$

This is a soft prior, not a hard equality constraint.

1.8 Optional cross-configuration coefficient regularization

The shared network weights already transfer information across configurations. If stronger configuration continuity is desired, one may add an explicit penalty on campaigns that share at least one sensor.

Let $\mathcal{P}$ be a set of campaign pairs (m, m') such that

$$\mathcal{S}_m \cap \mathcal{S}_{m'} \neq \varnothing.$$

For each pair, define a configuration kernel

$$K(c_m, c_{m'}) = \exp\left(-\frac{\|D_c(c_m - c_{m'})\|_2^2}{\tau_c^2}\right), \tag{28}$$

where D_c rescales the configuration coordinates into comparable units and $\tau_c > 0$ is a bandwidth parameter.

Then one may penalize coefficient discrepancies for sensors observed in both campaigns:

$$\mathcal{R}_{\text{cfg}} = \sum_{(m,m')\in\mathcal{P}} K(c_m, c_{m'}) \sum_{s\in\mathcal{S}_m\cap\mathcal{S}_{m'}} \left[\alpha_\gamma \|a^\gamma_{m,s} - a^\gamma_{m',s}\|_2^2 + \alpha_\eta \|a^\eta_{m,s} - a^\eta_{m',s}\|_2^2 + \alpha_\nu \|a^\nu_{m,s} - a^\nu_{m',s}\|_2^2\right]. \tag{29}$$

This term is optional. It encodes the prior that nearby configurations should induce nearby calibration curves for the same hardware, without forcing exact equality across configurations.

1.9 Global training objective

The global objective is

$$\mathcal{L}(\theta) = \frac{1}{M_{\text{camp}}} \sum_{m=1}^{M_{\text{camp}}} \mathbb{E}_{\mathcal{V}_m}\left[\mathcal{L}^{(m)}_{\text{mask}} + \lambda_{\text{vis}}\mathcal{L}^{(m)}_{\text{vis}} + \mathcal{R}^{(m)}_{\text{smooth}} + \lambda_{\text{anchor}}\mathcal{R}^{(m)}_{\text{anchor}}\right] + \lambda_{\text{cfg}}\mathcal{R}_{\text{cfg}} + \lambda_{\text{wd}}\|\theta\|_2^2. \tag{30}$$

Here:

- the expectation over $\mathcal{V}_m$ denotes random visible/masked sensor splits during stochastic training;
- $\lambda_{\text{vis}}, \lambda_{\text{anchor}}, \lambda_{\text{cfg}}, \lambda_{\text{wd}} \geq 0$ are scalar hyperparameters;
- if a term is not used, its coefficient is set to zero.

If external reference calibrations are available for some campaigns, a supervised term may be added. For example, if reference corrected measurements $X^{\mathrm{ref}}_{m,s,k,j}$ are available, one may augment (30) by

$$\mathcal{L}_{\mathrm{sup}} = \sum_{m\in\mathcal{M}_{\mathrm{sup}}} \sum_{s\in\mathcal{S}_m} \sum_{k=1}^{K_m} \sum_{j=1}^{J_m} \left(\widehat{X}_{m,s,k,j} - X^{\mathrm{ref}}_{m,s,k,j}\right)^2, \tag{31}$$

with an additional multiplier $\lambda_{\mathrm{sup}} \geq 0$.

1.10 Inference on a new campaign

Let

$$\mathcal{C}_\dagger = (\mathcal{S}_\dagger,\, K_\dagger,\, f_\dagger,\, c_\dagger,\, Y_\dagger)$$

be a new campaign. At inference time all available sensors are visible:

$$\mathcal{V}_\dagger = \mathcal{S}_\dagger.$$

The trained model produces

$$\{G_{\dagger,s,j}, N_{\dagger,s,j}, \sigma^2_{\dagger,s,j}\}_{s\in\mathcal{S}_\dagger,\, j=1,\ldots,J_\dagger}$$

through equations (4)–(17), then computes

$$\widehat{S}_{\dagger,k,j} = \left[\frac{\sum_{s\in\mathcal{S}_\dagger} \omega_{\dagger,s,j} G_{\dagger,s,j}\big(Y_{\dagger,s,k,j} - N_{\dagger,s,j}\big)}{\sum_{s\in\mathcal{S}_\dagger} \omega_{\dagger,s,j} G^2_{\dagger,s,j}}\right]_+,$$

and finally returns

$$\widehat{X}_{\dagger,s,k,j} = \frac{\big[Y_{\dagger,s,k,j} - N_{\dagger,s,j}\big]_+}{G_{\dagger,s,j}}.$$

Hence the trained system is a single global amortized model that maps

$$(Y_\dagger, \mathcal{S}_\dagger, f_\dagger, c_\dagger) \longmapsto (\widehat{X}_\dagger, \widehat{S}_\dagger, G_\dagger, N_\dagger, \sigma^2_\dagger),$$

while still preserving an explicit and interpretable calibration operator.

1.11 Interpretation

The model should be read as follows.

1. There is one global parameter vector θ shared across all campaigns.

2. For each new campaign, the network produces campaign-specific calibration parameters conditioned on the campaign measurements, the participating sensors, and the acquisition configuration.

3. The latent common spectrum is then recovered by an explicit weighted least-squares consensus step rather than by an unconstrained black-box decoder.

4. The returned calibrated PSD tensor is obtained by subtracting the inferred additive floors and dividing by the inferred multiplicative gains.

Thus the model is simultaneously:

- *configuration-conditional*, because the generated calibration depends on c_m;
- *campaign-specific*, because the outputs are inferred separately for each campaign;
- *amortized*, because one shared network replaces fitting a completely separate optimization model from scratch for every campaign;
- *hybrid and interpretable*, because the forward calibration equation remains explicit.